

Sex Hormone Involvement in the Induction of Experimental Systemic Lupus Erythematosus by a Pathogenic Anti-DNA Idiotypic in Naive Mice

MIRI BLANK, SHLOMO MENDLOVIC, HAROLD FRICKE, EDNA MOZES, NORMAN TALAL, and YEHUDA SHOENFELD

Abstract. The effect of sex hormones on the induction of experimental systemic lupus erythematosus (SLE) with a human anti-DNA (16/6 Id+) antibody, was studied. We found that injection of the pathogenic idiotype to BALB/c females and orchietomized males treated with estrogen caused a rapid outburst of the disease 3 months after immunization, while nonestrogen treated mice developed the disease 5 months after immunization. The flare of SLE disease was characterized by raised levels of autoantibodies in the sera to dsDNA, histones, cardiolipin, Sm, RNP, SSA (Ro), SSB (La) and an emergence of high titers of mouse antibody carrying the 16/6 Id. These enhanced antibody levels were associated with an increase in erythrocyte sedimentation rate, proteinuria and leukopenia. Immunofluorescent studies confirmed the existence of immune complexes in the afflicted kidneys. Testosterone treated BALB/c females and orchietomized males developed a classical response to the human anti-DNA antibody (16/6 Id+), but failed to develop fulminant SLE-like disease. Our data demonstrate the importance of sex hormones on the induction of experimental SLE-like disease in mice with no genetic tendency to autoimmunity. (*J Rheumatol* 1990;17:311-7)

Key Indexing Terms:

SYSTEMIC LUPUS ERYTHEMATOSUS

ANTI-DNA ANTIBODY IDIOTYPE

17 β ESTRADIOL

TESTOSTERONE

AUTOIMMUNITY

AUTOIMMUNE DISEASE

Sex steroid involvement in autoimmune diseases has been well studied in animals as well as in humans^{1,2}. The involvement of sex steroid hormones in systemic lupus erythematosus (SLE) was shown in (NZB $\times$ NZW) F₁, MRL-1pr/1pr and MRL+/+ mice, strains which develop spontaneous SLE³⁻⁶. Similarly androgens reduced, while estrogens enhanced, the spontaneous antibody production in mixed strains such as NZB $\times$ CBA and NZB $\times$ C3H mice^{7,8}. In the MRL/1pr mice, testosterone treatment reduced the lupus-like symptoms without affecting lymphoproliferation⁹. The role of hormones in autoimmunity was also demonstrated in human studies¹⁰⁻¹⁴.

We recently established an experimental system of SLE induction in mice^{15,16} employing a pathogenic idiotype of anti-DNA antibody (16/6 Id)¹⁶⁻¹⁹. This murine model for

SLE is now employed to study the effect of sex hormones on the earlier stages and development of SLE, in mice having no genetic tendency to develop SLE.

MATERIALS AND METHODS

Mice. BALB/c female and male mice were obtained from the Animal Resources, Sackler Medical School, Tel Aviv University, Israel. The mice were used at the age of 2 months.

Orchiectomy. The testes were exposed through a scrotal incision. The vas deferens and spermatic vessels were transected using electrocautery and the testes and epididymis were removed according to a described method³.

Hormone implantation. To achieve a constant hormone level in the females and orchietomized males we implanted the hormone as described³: silastic tubes (0.5) containing 2-3 mg of either 17 β estradiol powder (Sigma Inc.) or testosterone powder (Sigma Inc.) were implanted subcutaneously between the ears immediately after the orchiectomy. This treatment according to description resulted in an adequate hormone level within 12 h, which persisted for at least 3 months³. Control (placebo) mice were given empty implants.

Antibodies with the 16/6 idiotype. The human hybridoma secreting the antibody with the idiotype 16/6 (IgM) was grown in culture. The immunoglobulins were precipitated from the culture fluid with 50% (w/v) ammonium sulfate. The affinity purified material that was eluted from a goat antihuman IgM sepharose column was employed for immunization. The characteristics of the 16/6 antibody were described and summarized elsewhere¹⁶⁻²⁰. Briefly, it binds DNA, poly(I) and poly(G) and is deposited in affected skin and kidney of patients with SLE^{19,20}.

BR2 is another human monoclonal IgM generated by the hybridoma technique from lymph node lymphocytes of a patient with breast cancer. It was purified in the same manner as the 16/6 idiotype. The antibody does not bind DNA and does not carry the 16/6 Id.

From the Research Unit of Autoimmune Diseases, Department of Medicine B, Sheba Medical Center, Tel-Hashomer and Faculty of Medicine, Tel-Aviv University, the Department of Chemical Immunology, The Weizmann Institute of Sciences, Rehovot, Israel, and the Clinical Immunology Section, Department of Medicine, The University of Texas Health Sciences Center, San Antonio, TX, USA.

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M. Blank, PhD, Tel-Aviv University; S. Mendlovic, MD, PhD; H. Fricke, MSc; E. Mozes, PhD, Weizmann Institute of Sciences; N. Talal, MD, University of Texas Health Sciences Center; Y. Shoenfeld, MD, Tel-Aviv University.

Address reprint requests to Dr. Y. Shoenfeld, Department of Medicine 'B', Sheba Medical Center, Tel-Hashomer, 52629, Israel.

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Immunization procedure. One week after hormonal manipulation, several groups of BALB/c female and orchiectomized male mice were immunized with 1 μ g of the idiotype 16/6 or with BR2 in complete Freund's adjuvant (CFA; Difco) injected intradermally into the hind footpads as was described¹⁵.

Three weeks later booster injections were administered with 1 μ g of the idiotype 16/6 or with BR2 in isotonic phosphate buffered saline (PBS) in the hind footpads.

The following groups (5 mice in each group) were immunized: females; females transplanted with 17 β estradiol (Sigma Inc.); females implanted with testosterone (Sigma Inc.) or females transplanted with placebo; orchiectomized males; orchiectomized males treated with either 17 β estradiol, testosterone or placebo.

Antigens. The synthetic branched polypeptide (T,G)-A-L [poly(L-Tyr,L-Glu)-poly-(DL-Ala)-poly-(L-Lys)] served as a control antigen for detecting irrelevant antibodies in the experiment.

Autoantibodies detection by ELISA. Antibodies to ssDNA, dsDNA, poly(I), poly(G), histones (H1-5 subfractions), Sm, RNP, SSA(Ro), SSB(La) cardiolipin and 16/6 Id were tested as described^{15,17,21-23}.

Detection of SLE associated manifestations. The erythrocyte sedimentation rate (ESR) was determined by diluting heparinized blood in PBS at a 1:1 ratio. The diluted blood was passed to a microsampling pipette, and the sedimentation was measured 6 h later. Proteinuria was measured semiquantitatively by using a Combistix kit (Ames, Elkhart, IN)¹⁵. White blood cell count was performed utilizing Coulter counter (Estrogins Inc.).

Immunohistochemistry. Kidneys were removed and frozen immediately in liquid nitrogen. Five micrometer frozen cryostat sections were dried and fixed in acetone for 10 min. For the detection of Ig deposits FITC conjugated goat antimouse immunoglobulin antibodies were applied for 30 min. After extensive washing in PBS, the specific binding was visualized by a fluorescent microscope.

RESULTS

Estrogen effect on the induction of SLE in naive mice. Figure 1 summarizes the titers of the autoantibodies as determined in the groups of animals treated with 17 β estradiol, 3 months after immunization with 16/6 Id. High titers of anti-16/6 Id, which reached the maximum levels 2-3 weeks after the boost, were noted in the sera of the 16/6 Id immunized mice. The most remarkable autoantibody expression was observed in estrogen treated orchiectomized males immunized with 16/6 Id. Slightly lower levels of autoantibodies could be demonstrated in females treated with estrogen and immunized with 16/6 Id. The later titers were higher than those observed in females immunized with the pathogenic idiotype in which estrogen was not added. In both cases of estrogen treatment before immunization, the clinical and serological manifestations were more prominent (e.g., increased ESR, proteinuria, low WBC count). The high titers of antibodies in the 16/6 Id immunized mice remained high even 9 months after the immunization (groups of females, females treated with estrogen, and in orchiectomized males exposed to estrogen).

The levels of autoantibodies in the sera of orchiectomized males immunized with 16/6 Id reached their highest levels 2 months after immunization and then decreased towards the 5th month. The mice immunized with BR2 (IgM) developed a regular response to the foreign antigen and developed neither the clinical manifestations typical of SLE nor increased levels of autoantibodies. In addition, nonimmunized

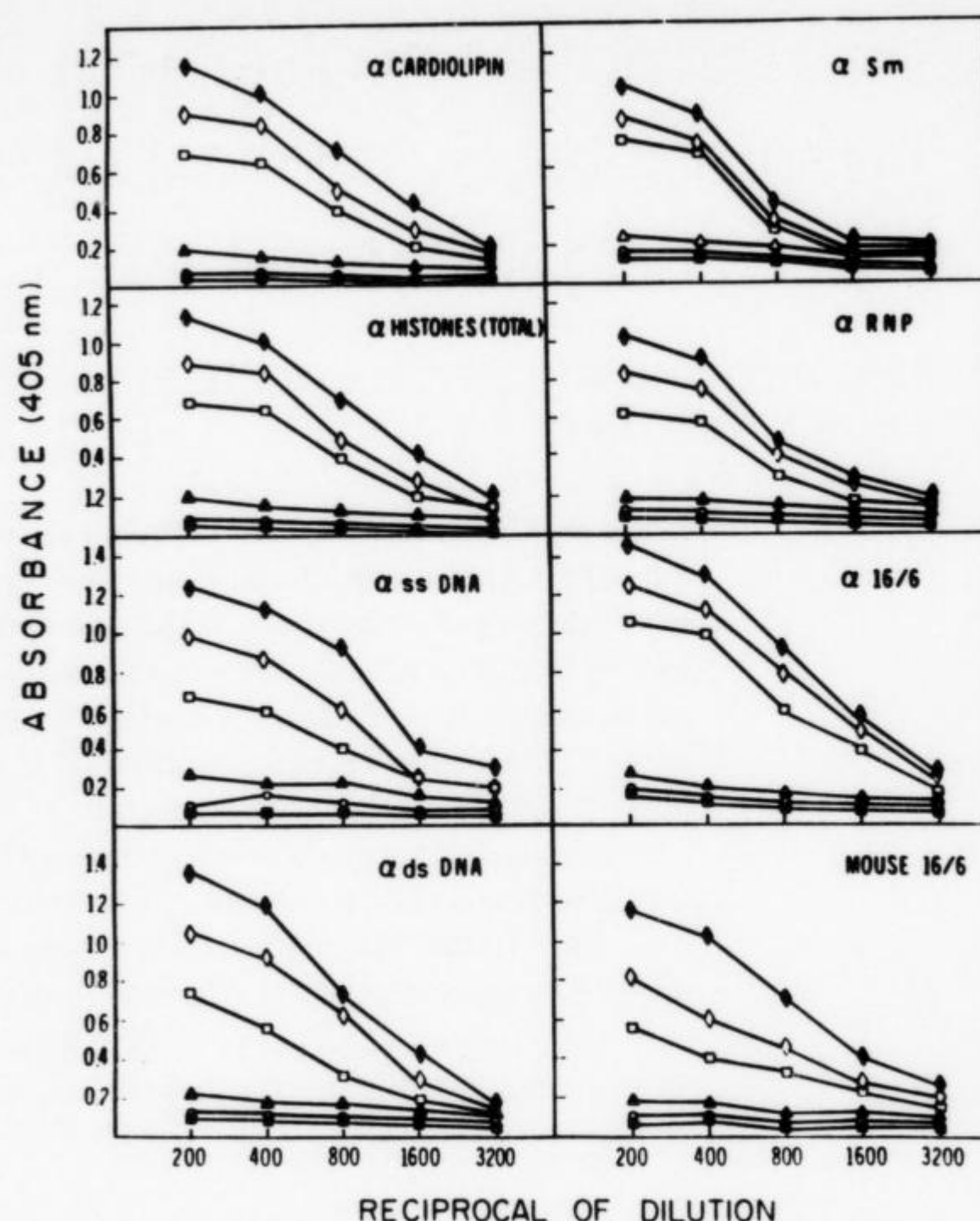


Fig. 1. Estrogen effect on the levels of autoantibodies units (of OD₄₀₅ nm) in sera of 16/6 Id immunized mice, treated with estrogen 3 months after injection (mean of 5 mice). $\blacklozenge$ — $\blacklozenge$ orchiectomized males immunized with 16/6 Id treated with 17 β estradiol; $\blacksquare$ — $\blacksquare$ orchiectomized males immunized with 16/6 Id and implanted with placebo; $\blacksquare$ — $\blacksquare$ orchiectomized males implanted with placebo; $\diamond$ — $\diamond$ females immunized with 16/6 Id and exposed to 17 β estradiol; $\circ$ — $\circ$ females implanted with placebo; $\square$ — $\square$ females immunized with 16/6 Id and implanted with placebo.

females and orchiectomized males, estrogen or placebo implanted, served as negative controls. In contrast, to the above mentioned mice, all the 16/6 Id immunized females and orchiectomized males that were treated with estrogens demonstrated enhancement in the titers of autoantibodies against the panel of autoantigens examined (Figure 1). The results summarized in Table 1 indicate that the elevated titers of antihistone antibodies seen in mice treated with estrogen 3 months after immunization with 16/6 Id were mainly directed towards the H1 and H2B subfractions. Lower titers were demonstrated in the sera of mice which were immunized with 16/6 Id but were not treated with estrogen.

Later in the course of the experiment (5 months) we could demonstrate an increase in the titers of autoantibodies in the sera of females, immunized with 16/6 Id (Table 2). The titers of autoantibodies in mice immunized with 16/6 Id and treated with estrogen reached peak levels 2 months after the immunization and remained high 5 months after immunization (Table 2). The enhanced levels in the titers of autoantibodies remained so, even 9 months after the immunization. No increased titers of antibodies were recorded against the syn-

Table 1. Antibodies against histone subfractions in mice treated with 17 β estradiol or testosterone and immunized with 16/6 Id

Mice	Immunization	Antihistone Antibodies Profile OD405 units $\times 10^3$					
		16/6	BR2 H1	H2-A	H2-B	H3	H4
POM	-		137 $\pm$ 7	104 $\pm$ 3	104 $\pm$ 11	79 $\pm$ 5	68 $\pm$ 4
PF	-		121 $\pm$ 8	111 $\pm$ 7	84 $\pm$ 4	73 $\pm$ 5	72 $\pm$ 3
POM	+		178 $\pm$ 8	122 $\pm$ 9	127 $\pm$ 5	130 $\pm$ 8	144 $\pm$ 11
PF	+		513 $\pm$ 11	227 $\pm$ 12	414 $\pm$ 12	212 $\pm$ 9	190 $\pm$ 7
EOM	-		122 $\pm$ 4	113 $\pm$ 5	119 $\pm$ 7	99 $\pm$ 4	104 $\pm$ 3
EOM	+		1126 $\pm$ 23	339 $\pm$ 16	843 $\pm$ 15	426 $\pm$ 12	642 $\pm$ 19
TOM	-		103 $\pm$ 4	68 $\pm$ 5	114 $\pm$ 9	87 $\pm$ 7	72 $\pm$ 4
TOM	+		121 $\pm$ 11	113 $\pm$ 8	94 $\pm$ 5	88 $\pm$ 4	109 $\pm$ 9
EF	-		79 $\pm$ 3	93 $\pm$ 5	101 $\pm$ 9	116 $\pm$ 7	110 $\pm$ 7
EF	+		878 $\pm$ 14	285 $\pm$ 12	752 $\pm$ 18	440 $\pm$ 9	303 $\pm$ 15
TF	-		93 $\pm$ 5	65 $\pm$ 3	96 $\pm$ 7	81 $\pm$ 5	71 $\pm$ 7
TF	+		114 $\pm$ 12	93 $\pm$ 8	127 $\pm$ 9	117 $\pm$ 7	102 $\pm$ 9
POM	-	+	75 $\pm$ 5	67 $\pm$ 5	111 $\pm$ 7	103 $\pm$ 3	83 $\pm$ 7
EOM	-	+	94 $\pm$ 4	82 $\pm$ 8	93 $\pm$ 6	66 $\pm$ 9	109 $\pm$ 3
PF	-	+	104 $\pm$ 8	91 $\pm$ 9	121 $\pm$ 8	98 $\pm$ 7	111 $\pm$ 11
EF	-	+	73 $\pm$ 12	52 $\pm$ 8	89 $\pm$ 5	77 $\pm$ 7	61 $\pm$ 4

The values demonstrate titers of antihistone antibodies in OD 405 nm $\times 10^{-3}$, mean of 5 mice, serum dilution 1:200, 3 months after immunization. P - placebo implanted; O - orchiectomized; M - males; F - females; E - estrogen treated; T - testosterone treated.

Table 2. Autoantibody levels in sera of mice treated with 17 β estradiol before immunization with human 16/6 Id

Antibody To:	EOM	POM* + 16/6	EOM** + 16/6	EOM + BR2	EF	PF† + 16/6	EF‡ + 16/6	EF + BR2
ssDNA	68 $\pm$ 4	285 $\pm$ 6	1,201 $\pm$ 15	103 $\pm$ 21	62 $\pm$ 5	703 $\pm$ 22	910 $\pm$ 12	162 $\pm$ 13
dsDNA	58 $\pm$ 3	397 $\pm$ 8	1,127 $\pm$ 17	112 $\pm$ 8	69 $\pm$ 3	1,113 $\pm$ 19	1,210 $\pm$ 12	163 $\pm$ 32
poly (I)	101 $\pm$ 11	398 $\pm$ 14	1,149 $\pm$ 19	129 $\pm$ 11	112 $\pm$ 8	993 $\pm$ 14	1,118 $\pm$ 16	141 $\pm$ 11
poly (G)	65 $\pm$ 4	369 $\pm$ 12	1,294 $\pm$ 17	147 $\pm$ 21	74 $\pm$ 3	998 $\pm$ 18	1,164 $\pm$ 17	159 $\pm$ 37
Histones	52 $\pm$ 3	166 $\pm$ 18	1,301 $\pm$ 21	101 $\pm$ 16	24 $\pm$ 15	1,119 $\pm$ 14	1,284 $\pm$ 17	98 $\pm$ 21
Cardiolipin	21 $\pm$ 12	189 $\pm$ 12	913 $\pm$ 16	97 $\pm$ 13	47 $\pm$ 16	649 $\pm$ 12	818 $\pm$ 14	103 $\pm$ 17
Sm	74 $\pm$ 4	139 $\pm$ 19	1,212 $\pm$ 19	134 $\pm$ 22	112 $\pm$ 12	794 $\pm$ 12	1,103 $\pm$ 14	127 $\pm$ 12
RNP	88 $\pm$ 5	135 $\pm$ 13	1,248 $\pm$ 16	149 $\pm$ 17	129 $\pm$ 9	909 $\pm$ 15	1,199 $\pm$ 18	135 $\pm$ 11
SS-A (Ro)	47 $\pm$ 5	164 $\pm$ 24	1,109 $\pm$ 15	155 $\pm$ 18	99 $\pm$ 8	665 $\pm$ 18	901 $\pm$ 17	129 $\pm$ 23
SS-B (La)	57 $\pm$ 4	177 $\pm$ 26	919 $\pm$ 13	103 $\pm$ 12	108 $\pm$ 9	805 $\pm$ 16	787 $\pm$ 18	139 $\pm$ 21
16/6	62 $\pm$ 7	843 $\pm$ 12	1,371 $\pm$ 27	47 $\pm$ 9	53 $\pm$ 6	1,479 $\pm$ 23	1,301 $\pm$ 14	131 $\pm$ 17
anti-16/6§	41 $\pm$ 3	187 $\pm$ 19	901 $\pm$ 17	45 $\pm$ 12	25 $\pm$ 4	579 $\pm$ 11	777 $\pm$ 18	37 $\pm$ 11

Antibody responses expressed as mean $\pm$ SD of sera taken from 5 mice, 5 months after immunization. Serum was diluted 1:200 and the results are presented as OD₄₀₅ nm $\times 10^3$. § Antibody to anti-16/6 represents the mouse 16/6 Id. * POM + 16/6 - placebo implanted orchiectomized males, immunized with 16/6 Id. **EOM + 16/6 - estrogen treated orchiectomized males, immunized with 16/6 Id. †PF + 16/6 - placebo implanted females immunized with 16/6 Id. ‡EF + 16/6 - estrogen treated females immunized with 16/6 Id.

thetic polypeptide (T,G)-A-L in all mice. Similarly, no auto-antibodies were detected in the sera of either estrogen treated or untreated unimmunized mice, or when immunized with the control antibody. Clinical manifestations such as increased ESR, proteinuria and leukopenia 3 months after immunization are shown in Table 3.

Testosterone effect on the induction of SLE in naive mice. The following groups of BALB/c mice were treated with testosterone then immunized with the 16/6 Id: (1) females immunized with 16/6 Id or BR2 were treated with testoster-

one or placebo; (2) orchiectomized males immunized with 16/6 or BR2 and treated with testosterone or placebo; and (3) females immunized with 16/6 Id and orchiectomized males treated with 17 β estradiol and then immunized with 16/6 Id served as positive controls.

The appearance of anti-16/6 or anti-BR2 antibodies could be seen as early as 2 weeks after the boost injection. It should be noted that the titers of anti-16/6 Id were lower in the testosterone treated mice. The high levels of anti-16/6 antibodies remained for 7 months only in the females group immunized with 16/6 Id or orchiectomized males estrogen

Table 3. Clinical findings in mice treated with estrogen and then immunized with 16/6 Id

	Groups (5 in each)	ESR (mm/h)	WBC (cells/mm)	Proteinuria (g/l)
Females	Untreated	0.75	5194 ± 958	<0.3
	Placebo implanted	1	4773 ± 955	<0.3
	Estrogen treated	2	4799 ± 863	<0.3
	16/6 immunized, Placebo implanted	3	4271 ± 691	<0.3
	16/6 immunized			
	Estrogen treated	9	3432 ± 970	<u>2</u>
Males	Untreated	0.8	4039 ± 993	<0.3
	Orchiectomized placebo implanted	1.5	4566 ± 670	<0.3
	Orchiectomized estrogen treated	2	4039 ± 840	<0.3
	16/6 immunized	2	4134 ± 1013	<0.3
	orchiectomized			
	16/6 immunized, orchiectomized			
	estrogen treated	10	2973 ± 784	<u>3</u>

The data represents ESR, WBC count and proteinuria 3 months after immunization.

treated, while mice treated with testosterone showed low titers of anti-16/6 similar to those recorded on males immunized with 16/6 or BR2. Slight elevation of antinuclear antibodies (ANA) was noted 2 months after the injection with a trend toward reduction in the upcoming months (Table 4).

The autoantibody pattern in orchiectomized males and females immunized with 16/6 and estrogen treated was parallel to that described (Table 1, Figure 1). Similar results were noted regarding the clinical variables. Neither anti-histones and cardiolipin antibodies nor clinical manifestations of disease could be found in the 16/6 immunized mice treated with testosterone.

Immunohistology of the kidneys in the different experimental groups. Figures 2, 3 demonstrate the immunohistology of kidney sections of mice belonging to the different experimental groups. As can be seen in females and males injected with the 16/6 Id, immunoglobulin deposits were found in the kidneys (Figures 2a, b). Estrogen treatment enhanced the immunoglobulin deposition in 16/6 Id injected males (Figure 2d), but had a less pronounced effect on females (Figure 2c). The administration of testosterone had a beneficial effect on kidney damage in males. Still, staining was observed in the kidney sections of testosterone treated females (Figure 3a), but no immunoglobulin deposits were detectable in the kidney of testosterone treated males (Figure 3b).

DISCUSSION

Our study suggests that sex hormones affect the induction of experimental SLE after immunization of BALB/c mice with the pathogenic human anti-DNA idiotype 16/6 Id. It should be emphasized that this strain of mouse does not develop any autoimmune disease spontaneously. The effects of sex hormones were shown in several experiments: (1) females treated with estrogen before immunization with 16/6 Id showed an accelerated experimental SLE induction when compared to control females immunized with 16/6 Id only; (2) testosterone treatment of mice before 16/6 Id immunization (females and orchiectomized males) resulted in a milder disease on both serological (autoantibody production) and clinical levels (increased ESR, proteinuria and leukopenia) levels.

The histological findings indicate that testosterone administration in immunized, orchiectomized males could prevent, while estrogen administration led to an earlier onset and more pronounced, kidney damage. Our data confirm reports showing that both androgens and estrogens can modify the expression of autoimmunity in mice which spontaneously develop

Table 4. Autoantibody profile in sera of mice immunized with 16/6 Id and supplemented with testosterone

Antibody To:	TOM	POM + 16/6	TOM + 16/6	TOM + BR2	EOM + 16/6	PF + 16/6
ssDNA	37 ± 5	212 ± 21	384 ± 27	73 ± 4	1,314 ± 57	849 ± 24
dsDNA	25 ± 3	228 ± 38	412 ± 29	67 ± 5	1,202 ± 44	937 ± 72
poly (I)	41 ± 5	233 ± 49	372 ± 56	45 ± 4	1,309 ± 52	798 ± 32
poly (G)	38 ± 7	199 ± 17	333 ± 48	53 ± 5	1,284 ± 67	784 ± 21
Histones	42 ± 4	53 ± 3	46 ± 7	51 ± 3	1,237 ± 71	675 ± 33
Cardiolipin	35 ± 5	64 ± 7	78 ± 4	62 ± 4	799 ± 23	513 ± 18
Sm	44 ± 6	48 ± 3	54 ± 5	46 ± 6	1,094 ± 45	643 ± 24
RNP	35 ± 4	56 ± 17	82 ± 7	69 ± 5	1,188 ± 69	534 ± 43
SSA (Ro)	27 ± 6	49 ± 5	71 ± 5	57 ± 9	979 ± 37	488 ± 78
SSB (La)	53 ± 3	67 ± 7	59 ± 6	72 ± 8	885 ± 35	404 ± 55
16/6	22 ± 5	868 ± 72	575 ± 38	33 ± 4	1,291 ± 98	1,024 ± 127
Anti-16/6	39 ± 4	127 ± 28	79 ± 5	23 ± 3	842 ± 80	679 ± 74

Antibodies level at OD 405 nm units × 10³. The titers were determined at 3 months after immunization in a sera dilution 1:200 (mean 5-7 mice in each group POM, TOM, EOM, FP, see Table 2.) Sera of orchiectomized males, nonimmunized or immunized with BR2, had no elevated levels of autoantibodies. TOM had no elevated titers until 9 months after immunization.

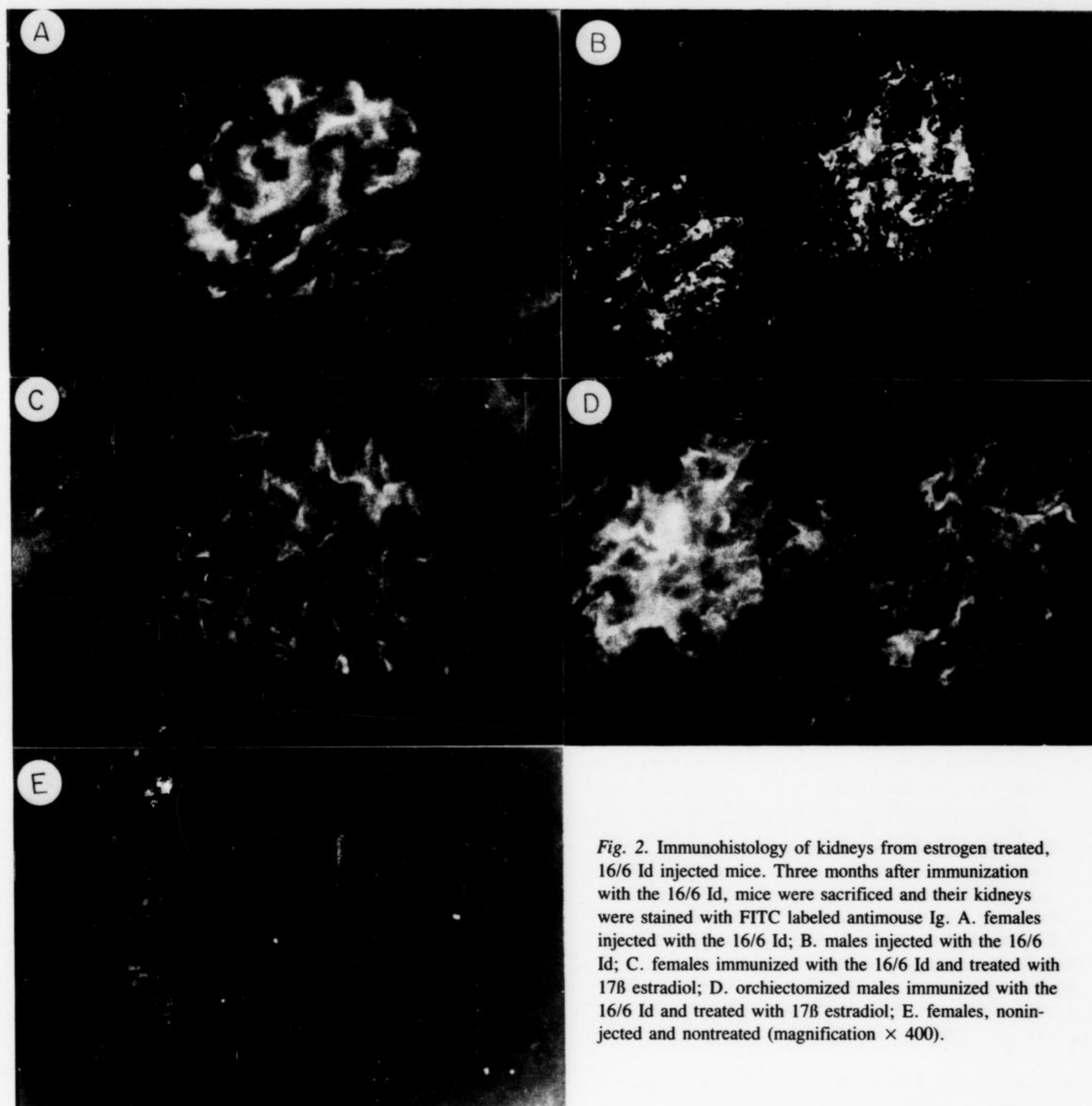


Fig. 2. Immunohistology of kidneys from estrogen treated, 16/6 Id injected mice. Three months after immunization with the 16/6 Id, mice were sacrificed and their kidneys were stained with FITC labeled antimouse Ig. A. females injected with the 16/6 Id; B. males injected with the 16/6 Id; C. females immunized with the 16/6 Id and treated with 17 β estradiol; D. orchiectomized males immunized with the 16/6 Id and treated with 17 β estradiol; E. females, noninjected and nontreated (magnification $\times$ 400).

severe SLE like disease [NZB/NZW F₁, NZB $\times$ CBA, NZB $\times$ C3H, MRL-1pr/1pr, MRL-+/+]^{6,9,13,24}.

The cellular mechanisms by which estrogen and testosterone affect the immune status of mice is still obscure. Several studies have suggested the important role of the thymus as a major target organ through which the sex hormone acts^{2,3,25-27}. Modulation of T cell mediated delayed type hypersensitivity reactions have been observed by sex hormones^{1,2,28}. Moreover, *in vivo* testosterone treatment reduced the lymphoproliferative response to synthetic protein derivative, a T cell antigen²⁸.

Androgens were also found to maintain suppressor cell activity and the number of Lyt-2 positive cells²⁹, while estrogens may suppress the reactivity of T cells²⁹⁻³¹.

Recently, we have shown that 16/6 Id induced experimental SLE can be prevented if the mice are treated with 16/6 Id T suppressor cells (Blank M, Shoenfeld Y, submitted for publication). It is therefore conceivable that the improvement noted with androgen and the deterioration observed with estrogen is mediated via their effects on T suppressor cells.

Alternatively, the production of cytokines by lymphocytes and macrophages is also affected by sex hormones. Thus, estrogen administration to mice increased the production of interleukin 1 (IL-1) and Ia expression³². Similarly androgens may stimulate increased IL-2 secretion, while estrogens induce hypoactivity of T cells^{30,31} and reduce IL-2 and gamma interferon production by T cells^{33,34}. Similar effects of estrogen on lymphokine generation were found in macro-



Fig. 3. Immunohistology of kidney from testosterone treated 16/6 injected mice. A. Females immunized with the 16/6 Id and treated with testosterone. B. Orchietomized males immunized with the 16/6 Id and treated with testosterone.

phages³⁵. Since reduced IL-2 production was reported to be a common feature in various autoimmune conditions³⁶, the opposite influence of estrogen and testosterone on this variable may affect the outcome of mice with experimental SLE. A third mechanism by which the sex hormones may mediate their effects in autoimmunity can be through the expression of Ia on immune as well as nonimmune cells. Testosterone treatment of normal and autoimmune mice profoundly reduced the number of Ia positive cells in the thymus and in mammary gland epithelium³⁷. It is not yet clear whether these effects, which may enhance or reduce antigen presentation, are due to change in levels of cytokines (e.g., interferon or tumor necrosis factor [TNF]) or are the result of a direct binding *per se*.

Sex hormones may also influence nonlymphoid cells which interact closely with lymphoid cells. For example, chemi-

cal bursectomy induced by testosterone is due to the effect of steroids on the epithelial cells, but not on the lymphoid cells³⁸ which might then affect the maturation and development of B cells. Involvement of steroid receptor protein as a target for sex hormone activity was suggested as well¹. It seems, therefore, that sex hormones may influence the immune system in many ways by affecting multiple organs and cells through mechanisms which are as yet not completely understood.

The serological findings in the 16/6 Id induced experimental SLE in mice is characterized by a "polyclonal activation" and increased levels of almost any ANA found in sera of patients with SLE. We presume that the strong stimulus by the pathogenic Id leads to the dysregulation of a large number of regulatory idiotypes that is expressed in an emergence of a great diversity of autoantibodies.

We have reported that sex hormones play a role in the development of SLE-like diseases after immunization with anti-DNA antibody carrying the 16/6 in naive mice having no genetic tendency. The manipulation of the induction of experimental SLE in naive mice by sex hormones may help to elucidate the role of these hormones in the pathogenesis of SLE.

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